



Administrative monopoly regulation and the cost of debt financing: Evidence from China

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ABSTRACT

Traditional antitrust research has primarily focused on the market-based monopoly, while less attention given to those formed by administrative power. This study explores how the Fair Competition Review System (FCRS) in China, designed to curb administrative monopoly, influences corporate debt financing costs. We find that administrative monopoly regulation reduces the cost of debt financing. Enhanced accounting information quality and lower corporate default risk are two possible channels. The main effect is more pronounced among private firms, small-sized and medium-sized firms, and firms in regions with lower degree of financial marketization. The FCRS also aids in addressing the issue of “zombie firms”, mitigating their crowding-out effect on the debt financing of normal firms. Moreover, the cost of debt financing decreases more when the review measures include both stringent self-inspection and external supervision. These findings underscore the importance of administrative monopoly regulation and contribute to the advancement of antitrust endeavors in the Asia-Pacific region.

1. Introduction

Government regulations aiming to cope with market-based or administrative-based monopolies are pervasive and important worldwide.¹ It is probable that monopoly regulation increases the debt financing costs of firms, as enhanced market competition can increase firm operating risk when the market size is unchanged (Valta, 2012). However, unlike general monopoly regulation, administrative monopoly regulation breaks the market segmentation and potentially expands the market size, which may be conducive to improving the operating performance of firms, and thus reduce their debt financing costs.² Since administrative monopoly characterized by local protection and market segmentation is widely present in countries around the world (Hasan et al., 2021), and debt financing is very important for firms, it is interesting and important to study the impact of administrative monopoly regulation on debt financing costs.

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¹ Recent literature on monopoly regulation has delved into both aspects (for example, Levine et al. (2021) and Ding et al. (2022) talk about market-based monopolies while Yang et al. (2024) studies administrative-based monopolies).

² Administrative monopoly refers to local governments restricting foreign firms from entering the local market through administrative means, thereby giving local enterprises a monopolistic position (Xu et al., 2023; Zhang et al., 2023). Administrative monopoly leads to market segmentation, and the products of firms can only be sold locally and are difficult to enter foreign markets. Administrative monopoly regulation restricts the local protectionist actions of the government, which is conducive to firms selling their products to a larger market.

Theoretically, administrative monopoly regulation could have both positive and negative effects on corporate debt financing costs. On the one hand, administrative monopoly regulation may reduce debt financing costs. Firstly, administrative monopoly regulation maintains fair market competition, which allows corporate performance to be intuitively evaluated by comparison with peers, making it difficult for management teams to conceal “bad news”, thereby improving the quality of corporate accounting information (Cheng et al., 2013; Guo et al., 2019) and reducing corporate debt financing costs (Sufi, 2007; Zhu et al., 2023). Secondly, increased market competition intensity stimulated by administrative monopoly regulation may encourage firms to improve productivity (Kong et al., 2022) and investment efficiency (Cho and Kang, 2019), thereby improving corporate financial performance and reducing corporate default risk, which in turn reduces corporate debt financing costs (Bolton and Scharfstein, 1990; Kang et al., 2016; Gigante and Manglaviti, 2022; Al Rabab’A et al., 2023).

On the other hand, administrative monopoly regulation may increase debt financing costs. Intensified market competition may weaken a firm’s competitive advantage, increase operational risk and damage profitability, thereby raising corporate debt financing costs (Hasan et al., 2021). Given that administrative monopoly regulation can affect corporate debt financing costs via different channels that yield opposite predictions, it is necessary to study this issue through empirical investigation.

We utilize a sample of Chinese A-share listed firms from 2014 to 2019 to investigate the impact of the Fair Competition Review System (FCRS), a regulatory framework introduced by the Chinese government to curb administrative monopoly, on corporate debt financing costs. There are three reasons for conducting this study in China. Firstly, the phenomenon of administrative monopoly in China is emblematic and merits scholarly attention. In China, because local government officials face with fierce competition when seeking for promotion, they have strong tendencies for local protectionism (Chen et al., 2005; Li and Zhou, 2005; Yao and Zhang, 2015). Local governments may shelter local firms and exclude outsiders by setting unreasonable and discriminatory access and exit conditions, implementing discriminatory pricing or subsidy policies, thereby granting local firms a certain monopoly status. The aforementioned measures cause a typical administrative monopoly phenomenon.

Secondly, in recent years, China has enacted a quintessential regulatory policy aimed at curbing administrative monopolies, which has garnered significant scholarly interest. The administrative monopoly phenomena existing in China distort factor prices, damage resource allocation efficiency, cause local protectionism and market segmentation, and hinder the free flow of goods and factors within China. To address these issues, the Chinese government has introduced a series of anti-administrative monopoly measures, the most important of which is the Fair Competition Review System (FCRS) established in June 2016 and implemented in various regions in 2017.

Finally, the allocation of credit resources in China is subject to a high degree of governmental intervention, which consequently renders the impact of administrative monopoly regulation on debt financing costs more pronounced. Due to the close connection between Chinese banks and local governments (in China, large banks are state-owned, and most local commercial banks are initiated by local governments), credit resources are subject to government control (Cull et al., 2017) and may be inclined towards local firms, thus forming an administrative monopoly phenomenon in corporate financing. At the same time, since debt financing, rather than equity financing, remains the main source of funding for Chinese firms (Gong et al., 2021), the cost of debt financing is important to firms and the whole economy.

Following previous literature on administrative monopoly (Xu et al., 2023; Zhang et al., 2023), we employ a difference-in-differences approach using the China’s implementation of the FCRS in 2017 as a quasi-natural experiment. The baseline regression results suggest that on average, the debt financing costs of treatment firms decreases by 0.355 %, resulting in a substantial reduction in interest expenses (19 million RMB on average). We confirm the robustness of our findings through multiple tests. The results of the mechanism analysis indicate that enhanced accounting information quality and lower corporate default risk are two possible channels. Furthermore, results of cross-sectional analyses suggest that the reduction in debt financing costs is more significant among private firms, small-sized and medium-sized firms, and firms in regions with lower degree of financial marketization. We also find that FCRS aids in addressing the issue of “zombie firms”, mitigating their crowding-out effect on the debt financing of normal firms. Additionally, when the review measures include both stringent self-inspection and external supervision, the effect of FCRS on reducing corporate debt financing costs is more significant.

The contributions of this paper can be summarized in three aspects. Firstly, this study enriches the literature on how administrative monopoly regulation affects firms. Early research has primarily focused on the macroeconomic and industry-level impacts of administrative monopolies, including macroeconomic hazards (Guo and Hu, 2004), determinants at the industry level (Dai et al., 2019) and regulatory challenges (Chan, 2009). In recent years, with the advancement of antitrust efforts in the Asia-Pacific region, especially in China, studies have begun to examine the impact of administrative monopoly regulation at the firm level, such as innovation (Xiao et al., 2023), zombie firm resolution (Fan et al., 2023) and cross-regional resource allocation (Zhao et al., 2024). However, no existing studies have examined the impact of administrative monopoly regulation on corporate debt financing costs. Thus, this study provides a new perspective for understanding the micro-level effects of administrative monopoly regulation comprehensively.

Secondly, we innovatively examine the impact of government policies on corporate financing from the perspective of regulating administrative monopolies, enriching the research on the determinants of corporate debt financing costs. Previous literature has shown that market-based monopoly regulation heightens market competition, increasing the demand for debt financing and thereby affecting debt costs (Giroud and Mueller, 2011; Valta, 2012; Huang and Lee, 2013; Boubaker et al., 2018; Hasan et al., 2021). However, existing literature rarely discusses the effect of administrative monopoly regulation on debt costs. Unlike market-based monopoly regulation, administrative monopoly regulation breaks market segmentation, expanding the potential market size for enterprises, which may be beneficial for improving corporate performance and thus reducing the cost of debt financing. We explore this interesting but unresolved issue, expanding the research on the determinants of corporate debt financing costs.

Thirdly, this study reveals the positive role of FCRS in reducing corporate debt financing costs, which not only provides strong empirical support for the in-depth development of the antitrust cause in the Asia-Pacific region but also offers an effective policy tool for policymakers in other transitional countries. Furthermore, we find that the policy effectiveness of FCRS is more pronounced when stringent self-inspection is combined with robust external supervision, providing valuable experience for other countries in formulating and implementing similar policies.

The rest of this paper is structured as follows. [Section 2](#) outlines the institutional context of FCRS and formulates our hypotheses. [Section 3](#) details the sample selection and research methodology. [Section 4](#) reveals the core findings of the empirical investigation. [Section 5](#) offers further analysis. [Section 6](#) presents the conclusion.

2. Institutional background and hypotheses development

2.1. Institutional background

Due to the existence of the “Chinese promotion tournament”, local governments in China are prone to using administrative measures to intervene in the economy, supporting local firms and excluding those from other regions. Specific measures include: coercing or subtly compelling operators to conduct transactions with certain local firms; excluding or restricting competition by entering into cooperation agreements or memoranda with certain local firms; hindering the free flow of goods between regions, such as establishing local protectionist barriers; excluding or limiting firms from other regions from participating in bidding and other business activities. These administrative measures grant certain local firms a monopolistic status, thus they are referred to as administrative monopolies.

To regulate monopolistic issues, China has successively introduced a series of antitrust laws and regulations, including the “Anti-Unfair Competition Law” and the “Anti-Monopoly Law”. However, the regulatory targets of these policies are all market-based monopoly and are difficult to effectively alleviate the administrative monopoly issues caused by governments’ intervening in the economy. To address administrative monopoly issues, the Chinese central government passed the decision to implement the first anti-administrative monopoly policy—the Fair Competition Review System (FCRS)—in June 2016. This policy was formulated into specific measures and implemented by local governments in 2017. Unlike previous antitrust policies, FCRS innovatively takes a range of policy measures that limit or exclude competition, formulated by policy-making bodies, as the object of regulation, with the goal of weakening administrative monopoly. Specifically, FCRS mainly targets policy documents pertaining to the economic behaviors of market entities formulated by administrative bodies and organizations authorized by laws and regulations to manage public affairs. The scope of the FCRS encompasses market entry and exit, industrial growth, investment attraction, bidding and tendering processes, government procurement, business operation standards, qualification criteria and so on. The review criteria mainly include four aspects, eighteen standards, and fifty-four scenarios. The 18 standards include five market access and exit standards (15 scenarios), five goods and factor free flow standards (21 scenarios), four standards affecting production and operating costs (10 scenarios) and four standards affecting production and operating behaviors (8 scenarios).

The FCRS has been significantly effective in combating administrative monopolies. According to statistics, from 2016 to 2020, the FCRS has reviewed 1.89 million documents in total pertaining to the economic behaviors of market entities, revising or abolishing nearly 30,000 documents; and reviewed 857,000 new policies, correcting more than 4100 policies that violate the review standards. Since 2022, a total of 163,700 incremental policy measures have been reviewed, 452,000 stock policies have been cleaned up, and 20,400 have been revised or abolished. The implementation of the FCRS has greatly alleviated China’s administrative monopoly issues.

As an important component of antitrust policy, the FCRS is of significant interest to the government, firms, policy researchers, and other stakeholders. Furthermore, since this system is established by the government, individual firms have little influence over its design and implementation. This setup offers an ideal quasi-natural experiment environment for investigating the link between administrative monopoly regulation and corporate debt financing costs. Consequently, we leverage FCRS as a quasi-natural experiment and utilizes a DID approach to assess the effects of administrative monopoly regulation policies on corporate debt financing costs. By examining the impact of China’s antitrust policies on corporate debt financing, we seek to offer insights that extend beyond the Chinese context. Our objective is to enrich the academic discourse on administrative monopoly regulation policies and corporate financing.

2.2. Hypotheses development

In China, banks typically have close ties with local governments. Since most large banks are state-owned and local commercial banks are often initiated by local governments, the government can exert considerable control over credit resources and influence their allocation ([Cull et al., 2017](#)). In order to win the regional competition and gain promotion, local governments’ officials may skew credit resources towards certain local firms, thereby creating an administrative monopoly phenomenon in corporate financing. These administrative monopoly measures distort the pricing of capital factors and impair the efficiency of credit resource allocation. The implementation of FCRS helps to alleviate administrative monopoly issues and may reduce corporate debt financing costs through the following two pathways.

The first pathway is the enhancement of accounting information quality. The implementation of the FCRS is conducive to enhancing the quality of corporate accounting information. On one hand, the FCRS can suppress the self-serving motives of corporate management teams, thereby improving the quality of corporate accounting information. Rent-seeking theory posits that governments’ intervention in the economy may induce rent-seeking activities ([Krueger, 1974](#)). Accordingly, government intervention under

administrative monopoly fosters rent-seeking opportunities, and the introduction of FCRS can effectively mitigate this issue. The FCRS prohibits local governments from enacting or implementing policies that affect the fair competition of market entities, reducing the space for corporate management rent-seeking and consequently decreasing their self-serving information manipulation behavior. At the same time, considering the significant impact of FCRS on firms with administrative monopoly status, the reform signals conveyed by FCRS can also attract the attention of external supervisors to these firms (Barber and Odean, 2008), increasing the cost of their management's misconduct and thus inhibiting their self-serving information manipulation behavior. This leads to an improvement in the accounting information quality of these firms, alleviating the information asymmetry between them and investors, and ultimately reducing their debt financing costs (Sufi, 2007).

On the other hand, the FCRS can also increase competitive pressure on firms, thereby forcing them to enhance the quality of their accounting information. By alleviating administrative monopoly issues, the FCRS has intensified the level of market competition to some extent. While fierce market competition increases firm risk, it can also have a certain corporate governance effect. Signaling theory suggests that as market competition intensifies, firms have an incentive to broadcast favorable information about themselves to distinguish themselves, and the accounting information quality, as an external manifestation of good internal control, is a common positive signal. Therefore, intense market competition will force firms to enhance their accounting information quality (Cheng et al., 2013; Burks et al., 2018; Guo et al., 2019), providing investors with more comprehensive and accurate firm information, and thus reducing the credit risk they face and the risk premium they demand (Zhu et al., 2023).

The second pathway is the reduction of corporate default risk. On one hand, the FCRS can improve a firm's operational and investment efficiency, thereby enhancing its financial performance and reducing the corporate default risk. The acquisition of an administrative monopoly status is not without cost. Administrative monopolistic firms often need to follow government instructions in production, investment and finance to maintain political connections with the government, which harms their efficiency. After the implementation of FCRS, firms no longer have to blindly follow government directives and can focus on maximizing shareholder wealth. At the same time, increased market competition may also stimulate firms to improve productivity (Kong et al., 2022) and investment efficiency (Cho and Kang, 2019), thereby improving their financial performance, reducing the corporate default risk, and consequently lowering their debt financing costs (Bolton and Scharfstein, 1990; Kang et al., 2016; Gigante and Manglaviti, 2022; Al Rabab'A et al., 2023).

On the other hand, the FCRS can improve a firm's profitability by reducing costs and increasing revenue, thereby reducing the corporate default risk. In the past, firms, despite having administrative monopoly status, often had to follow government directives in their procurement decisions. After the implementation of the FCRS, firms can freely choose the most favorable suppliers nationwide. At the same time, the implementation of the FCRS has broken down market segmentation, allowing products that were previously only sold locally to be sold throughout the country, expanding the potential market size for firms and increasing their product sales volume, thereby increasing revenue. The improvement in corporate profitability reduces the corporate default risk, and consequently, the interest rates demanded by creditors (Merton, 1974).

Building on the earlier analysis, we formulate Hypothesis H1a.

H1a. The implementation of the FCRS results in lower corporate debt financing costs.

However, the implementation of the FCRS may also increase the corporate debt financing costs through three channels. Firstly, after the implementation of the FCRS, firms that originally enjoyed administrative monopoly status can no longer receive preferential credit resources from the government and no longer possess special advantages such as subsidies from the government, tax breaks and implicit government guarantees (Pevzner et al., 2015; Dhaliwal et al., 2016), increasing the volatility of their future earnings and the corporate default risk. The increase in the risk of debt default will prompt creditors to demand a higher risk premium, thereby raising the cost of debt financing (Bolton and Scharfstein, 1990; Kang et al., 2016; Gigante and Manglaviti, 2022; Al Rabab'A et al., 2023; Ni and Xu, 2023).

Secondly, the FCRS weakens the competitive advantages of administrative monopoly firms, forcing them to face more intense competition. This may increase the volatility of their future cash flows, thereby raising their risk of default (Hasan et al., 2021).

Thirdly, the FCRS weakens creditors' perception of implicit guarantees, thereby increasing firms' debt financing costs. The government's implicit guarantees for administrative monopoly firms lead creditors to hold a rigid redemption expectation, which in turn causes them to underestimate the risk of debt default. For creditors, their perception of implicit government guarantees is shaped not only by objective facts but also by their own subjective judgments (Pevzner et al., 2015). With the continuous advancement of local government debt management reforms in China, the government has become increasingly cautious in the use of debt funds. Moreover, the frequent defaults on state-owned enterprise bonds have sent signals to creditors that implicit guarantees are being weakened. This has heightened creditors' attention to and sensitivity towards changes in implicit guarantees. Against this backdrop, the introduction of the FCRS further undermines creditors' perception of implicit guarantees, thereby increasing firms' debt financing costs.

Fourthly, the FCRS severs the government's financing support to firms, preventing them from obtaining funds at below-market prices through government intervention. This, in turn, increases firms' debt financing costs. In China, the majority of credit resources are directly or indirectly controlled by the government (Cull et al., 2017). Firms that enjoy administrative protection have been able to easily obtain preferential loans mandated by the government to banks, thereby securing external financing at very low interest rates. The FCRS cuts off the channels through which administrative power can expand into the economic sphere (Xu et al., 2023). As a result, administrative monopoly firms must now compete with other capital seekers for these credit resources and can no longer obtain credit at interest rates below the normal market price, thereby increasing firms' debt financing costs.

Considering these possibilities, this paper proposes a competing hypothesis H1b.

H1b. The implementation of the FCRS results in higher corporate debt financing costs.

3. Data and research methodology

3.1. Sample and data

Drawing upon the CSMAR (China Stock Market & Accounting Research) database, this study constructs a sample of Chinese A-share listed firms spanning the period from 2014 to 2019.³ After excluding financial firms and firms that are subject to special treatment (ST) or have incomplete data for essential variables, we arrive at a final sample of 10,272 firm-year observations covering 1712 firms.⁴ Continuous variables were winsorized at the 1st and 99th percentiles to curb the influence of extreme values.

3.2. Measures of variables

3.2.1. Dependent variable

Referring to Zhu et al. (2023), we adopt the ratio of interest expense to interest-bearing debt, scaled by 100, to measure the cost of debt (*COD*). Interest-bearing debt is defined as the sum of short-term borrowings, long-term borrowings, long-term liabilities due within one year, and bonds payable.

3.2.2. Independent variable

Our independent variable is a policy variable (*DID*). Given the nationwide implementation of FCRS, natural experimental and control groups are absent. However, regions with higher administrative monopoly are expected to experience a greater policy impact. We measure regional administrative monopoly using the degree of local government intervention in firms across Chinese provinces⁵ and group sample firms based on the average administrative monopoly level from 2014 to 2016. Firms in regions above the average are assigned to the experimental group, while those at or below the average are assigned to the control group. Following Zhao et al. (2024), we take the year 2017 as the policy implementation point since it was put into effect by the local governments in 2017. In summary, if the firm in the sample belongs to the experimental group and the year is 2017 or later, the *DID* value is 1; otherwise, it equals 0.

3.2.3. Control variables

Following Ni et al. (2022), we utilize a range of control variables ($Control_{i,t-1}$), which specifically include firm size (*Size*), ratio (*Lev*), firm age (*Age*), growth potential (*TobinQ*), profitability (*ROA*), cash flow ratio (*CFO*), and the level of local economic development (*lnGDP*). To reduce the potential risk of endogeneity, we lag the control variables by one period. The definitions for main variables are shown in Column (2) of Table 1.

3.3. Empirical design

We employ a DID approach and the model is presented as follows:

$$COD_{i,t} = \alpha + \beta DID_{j,t} + \gamma Controls + \varphi_{firm} + \tau_t + \varepsilon \quad (1)$$

where $COD_{i,t}$ represents the debt financing cost of firm i in year t , $DID_{j,t}$ is the policy variable corresponding to province j where firm i is located in year t , *Controls* is a set of control variables for firm i in year $t-1$, φ_{firm} denotes firm fixed effects, τ_t represents year fixed effects, and ε is the error term. Additionally, to enhance the robustness of the regression results, this paper clusters the robust standard errors at the provincial level.

4. Main empirical results

4.1. Descriptive statistics

Table 1 presents the descriptive statistics of main regression variables. The average cost of debt (*COD*) for Chinese A-share listed firms from 2014 to 2019 is 6.132 %, with the 1 % percentile at 0.300 % and the 99 % percentile at 38.727 %, indicating a significant variation in debt financing costs across different firms. The average value of the policy variable (*DID*) is 0.126, suggesting that firms in the experimental group, located in regions with a high degree of administrative monopoly, account for approximately 12.6 % of all firms.

³ Appendix A explains the reason for choosing this sample period.

⁴ Table A.1 in Appendix A illustrates the detailed sample selection process.

⁵ The marketization indexes of various provinces in China come from the China Market Index Database at: <https://cmi.ssap.com.cn/>.

Table 1

Variable definitions and descriptive statistics.

This table provides definitions and statistical characteristics of the main variables. The second column details the definitions and calculation methods for the main variables. The third to the last columns display their descriptive statistics.

Variable	Definition	N	Mean	Sd	p1	Median	p99
COD	$\text{Interest Expense} \div \text{Interest-Bearing Debt} \times 100$	10,272	6.132	4.884	0.300	5.286	38.727
DID	If the firm is located in a region with a higher degree of administrative monopoly and the year is 2017 or later, then the value is 1; otherwise, the value is 0.	10,272	0.126	0.332	0.000	0	1.000
lnGDPA	Ln(per capita GDP)	10,272	11.164	0.414	10.258	11.185	12.009
Age	Ln(years of existence)	10,272	19.737	5.254	8.000	20	33.000
Size	Ln(total assets)	10,272	22.716	1.318	20.329	22.539	26.660
Lev	Total liabilities $\div$ Total assets.	10,272	0.487	0.179	0.119	0.487	0.883
ROA	Return on total assets.	10,272	0.029	0.053	-0.243	0.028	0.159
TobinQ	Tobin's Q value.	10,272	1.886	1.119	0.870	1.541	6.877
CFO	Cash flow generated from operating activities $\div$ Total assets.	10,272	0.044	0.069	-0.146	0.044	0.232

4.2. Baseline regression results

Table 2 displays the regression outcomes based on Eq. (1) without and with control variables. In both cases, the coefficient for DID_t is significantly negative, indicating that the experimental group firms experienced a more pronounced reduction in debt financing costs compared to the control group. This supports Hypothesis H1a, suggesting that the FCRS significantly lowers corporate debt financing cost.

Specifically, Column (2) of Table 2 shows that the FCRS implementation is associated with a 0.355 % reduction in debt costs, equivalent to 4.5 % of the average debt financing cost of 7.833 %. For a firm with average interest-bearing debt of 5320 million RMB, the FCRS could help firms save up to 18.886 million RMB in interest expenses. These savings can remarkably enhance the firm's financial position, providing additional capital for operations and strategic investments, thereby promoting long-term growth and market competitiveness.

4.3. Robustness tests

4.3.1. Parallel trend test

The effective application of the DID method requires parallel trends. To examine the validity of the parallel trend assumption, we introduce a dummy variable $Policy_{y,i,t}$ on the basis of Eq. (1), constructing the model as shown in Eq. (2).

Table 2

Baseline regression results.

This table presents the baseline regression results. Column (1) and (2) report the regression results of COD on the policy variable DID without and with control variables, respectively. The values within parentheses represent the t-statistics and ***, **, * indicate statistical significance at the 1 %, 5 %, and 10 % level, respectively.

	(1)	(2)
$Dep=$	COD	COD
DID	-0.497*** (-2.858)	-0.355** (-2.310)
lnGDPA		0.027 (0.054)
Age		0.048 (0.443)
Size		0.023 (0.129)
Lev		7.761*** (15.002)
ROA		-0.310 (-0.683)
TobinQ		0.075 (1.384)
CFO		-2.096*** (-4.661)
_cons	0.900*** (44.645)	-4.172 (-0.615)
Firm	Yes	Yes
Year	Yes	Yes
N	10,272	10,133
R-squared	0.664	0.704

$$COD_{i,t} = \alpha + \sum_{y=2015}^{2019} \beta_y Policy_{y,i,t} + \gamma Controls + \varphi_{firm} + \tau_t + \varepsilon \quad (2)$$

where $Policy_{y,i,t}$ is a dummy variable. If firm i is in the experimental group in year t , and y equals the year corresponding to t , then $Policy_{y,i,t}$ equals 1; otherwise, it equals 0. The baseline period is used as the reference group, hence $Policy_{y,i,t}$ does not include $Policy_{2014,i,t}$.

Panel A of Table 3 displays the results of the parallel trend test. The estimated coefficients of $Policy_{2014}$ and $Policy_{2015}$ are not significant, which supports the parallel trend hypothesis and proves the robustness of our results.

4.3.2. Placebo test

To conduct a placebo test, we randomly select firms from the sample to define as the experimental group and then constructs a new policy variable DID_{fake} . To prevent other low-probability events from affecting the estimation results, we perform 1000 repetitions of random sampling. Fig. 1 presents the estimated coefficients' kernel density distribution for the randomly produced policy variable. The coefficients of DID_{fake} are essentially normally distributed with a mean of 0. The results show that the randomly generated policy variable does not significantly affect corporate debt financing costs, suggesting that the effect of FCRS on corporate debt financing costs is not driven by random factors, and the baseline regression results are reliable.

4.3.3. Propensity score matching

To find appropriate control group firms that are "similar" in key characteristics to the experimental group firms for a more accurate comparative analysis, we utilize the Propensity Score Matching (PSM) approach. Specifically, we select *Age*, *Size*, *Lev*, *ROA*, and *TobinQ* as matched variables and conduct a 1-to-2 closest neighbor matching with a caliper width set at 0.05. After matching, we firstly test the balance of the matched sample pairs, with the results displayed in Panel B of Table 3. It indicates that the mean differences between the control and experimental groups for each matched variable are not significant. This indicates that after propensity score matching, the differences between the control and experimental groups in observable characteristics are no longer significant.

After completing the Propensity Score Matching (PSM), we conduct regression analysis on the control and experimental groups with similar key characteristics using the same method as the baseline regression, with the results presented in Panel C of Table 3. The outcomes of the PSM-DID regression align with the baseline regression results, further confirming the role of FCRS in reducing corporate debt financing costs and effectively mitigating potential endogeneity issues that could interfere with the conclusions of this paper.

4.3.4. Other robustness tests

To validate the impact of FCRS implementation on corporate debt financing costs, we also conduct other robustness tests including replacing the independent and dependent variable, excluding relocating firms, using an unbalanced panel dataset and incorporating post-COVID-19 sample. Our results remain consistent under these tests and the detailed results are displayed in Appendix B.

5. Further analysis

5.1. FCRS and accounting information quality

Firstly, we examine whether the FCRS is conducive to improving the accounting information quality. Referring to Chan et al. (2009), we measure the accounting information quality with discretionary accruals (*DisAcc*). The regression results, as shown in Column (1) of Table 4, indicate that the coefficient for the policy variable *DID* is negative at the 5 % significance level, suggesting that the FCRS significantly reduced discretionary accruals (*DisAcc*), improved the accounting information quality, and thereby reduced the corporate debt financing costs.

5.2. FCRS and corporate default risk

Subsequently, we examine whether FCRS helps to reduce the corporate default risk. Our indicator for assessing corporate default risk utilizes a metric of expected default frequency (*EDF*), inspired by the approach outlined in Bharath and Shumway (2008). The regression results, as displayed in the Column (2) of Table 4, indicate that the coefficient for the policy variable *DID* is significantly negative, indicating that the enforcement of the FCRS has significantly reduced the expected default probability of firms, thereby reducing the corporate default risk and consequently lowering their debt financing costs.

5.3. Heterogeneity analysis

5.3.1. Firm ownership

Firstly, we perform an analysis of heterogeneity taking into account the ownership structure of firms. According to the ownership, firms are categorized into private firms (*Private*) and state-owned firms (*State-owned*), and subgroup regressions are performed. Columns (1) and (2) of Table 5 present the results of heterogeneity analysis based on ownership. The coefficient of *DID* is significantly negative for private firms, and it is negative but not significant for the state-owned firms. This finding indicates that the FCRS has a

Table 3

Parallel trend test and propensity score matching.

This table presents the results of parallel trend test and propensity score matching. Panel A displays the results of parallel trend test. Panel B presents the PSM (Propensity Score Matching) covariate balance between the matched pairs of FCRS and non-FCRS firms. Panel C presents the impact of FCRS on the corporate cost of debt after propensity-score matching. Due to space limitations, the regression coefficients for the control variables (*lnGDPA*, *Age*, *Size*, *Lev*, *ROA*, *TobinQ* and *CFO*) are not listed.

Panel A: Parallel trend test					
	(1)				
<i>Dep=</i>	COD				
Policy2014	0.237 (0.164)				
Policy2015	0.175 (0.126)				
Policy2017	−0.385** (0.159)				
Policy2018	−0.302* (0.149)				
Policy2019	−0.394* (0.205)				
Control	Yes				
Firm	Yes				
Year	Yes				
N	10,272				
R-squared	0.664				

Panel B: PSM covariate balance					
Matched variable	FCRS	NonFCRS	Bias (%)	t-Test	
				t-Statistics	p-Value
Age	21.472	21.564	−1.900	−0.480	0.635
Size	22.682	22.669	1.000	0.250	0.801
Lev	0.455	0.454	0.500	0.120	0.904
ROA	0.034	0.034	−0.100	−0.020	0.986
TobinQ	1.726	1.749	−1.900	−0.540	0.588

Panel C: Regression after propensity-score matching	
	(1)
<i>Dep=</i>	COD
DID	−0.568** (−1.711)
Control	Yes
Firm	Yes
Year	Yes
N	2604
R-squared	0.652

more pronounced effect on reducing the debt financing costs of private firms. This may be because the FCRS has weakened the administrative monopoly advantages that those state-owned firms previously enjoyed (Majumdar, 1998), resulting in a less significant reduction in debt financing costs compared to private firms.

5.3.2. Firm size

Secondly, we perform an analysis of heterogeneity taking into account the size of firms. We utilize the natural logarithm of total assets as a metric for assessing firm size, categorizing firms into small-sized and medium-sized firms (*SMF*) and large firms (*LF*) based on size, and performs subgroup regressions. Columns (3) and (4) of Table 5 present the results of the heterogeneity analysis based on firm size. The coefficient for the *SMF* group is significantly negative, while the coefficient for the *LF* group is negative but not

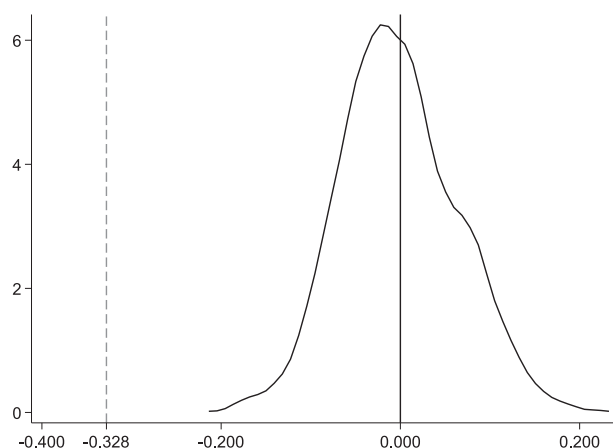


Fig. 1. Placebo test.

Notes: The horizontal axis represents the estimated coefficients of the randomly generated policy variable DID_{fake} , the curve represents the kernel density distribution, and the dashed line represents the estimated coefficient of DID_{fake} from the baseline regression.

Table 4

Possible channels.

This table shows the two possible channels through which the FCRS reduces the cost of debt financing. Column (1) presents the impact of FCRS on the accounting information quality. The dependent variable is the discretionary accruals ($DisAcc$). The lower the $DisAcc$, the higher the quality of accounting information for listed firms. Column (2) presents the impact of FCRS on the corporate default risk. The dependent variable is the expected default probability (EDP). The higher the expected default probability, the higher the corporate default risk.

	(1)	(2)
Dep=	DisAcc	EDP
DID	−0.018** (−2.113)	−0.255* (−1.809)
Control	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes
N	9956	10,109
R-squared	0.093	0.662

Table 5

Heterogeneity analysis based on ownership and size.

This table presents the results of heterogeneity analysis based on ownership and size. Columns (1) and (2) respectively present the regression results for the private firm group and the state-owned firm group. Columns (3) and (4) respectively display the regression results for the small and medium-sized firms (SMF) and large firms (LF).

	(1)	(2)	(3)	(4)
	Ownership		Size	
	Private	State-owned	SMF	LF
Dep=	COD	COD	COD	COD
DID	−0.313** (−2.139)	−0.485 (−0.625)	−0.506** (−2.090)	−0.119 (−1.171)
Control	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	5874	4259	5623	4510
R-squared	0.711	0.681	0.705	0.801

significant. This suggests that FCRS has a more pronounced effect on reducing the debt financing costs of $SMFs$. This may be because large firms employ more workers, and local governments, aiming to ensure employment stability, were more likely to support them using administrative power before the implementation of the FCRS. The FCRS has diminished the administrative monopoly advantages

that large firms previously enjoyed, resulting in a less significant reduction in debt financing costs compared to *SMFs*.

5.3.3. Financial marketization degree

Thirdly, we conduct an analysis taking into account the degree of financial marketization across various provinces in China. We use the secondary index “Financial Marketization Index” of the Marketization Index to measure the degree of financial marketization in different provinces. Based on the average degree of financial marketization across various provinces in China from 2014 to 2016, we divide the samples into two groups: high degree of financial marketization group (*High*) and low degree of financial marketization group (*Low*), and perform regressions. As presented in Columns (1) and (2) of Table 6, the coefficient of *DID* in the *Low* group is significantly negative, while the coefficient for the *High* group with a higher degree of financial marketization, although negative, is not significant. This suggests that the role of FCRS in decreasing corporate debt financing costs is more pronounced in firms located in regions whose financial marketization degree is lower. The reason may be that in regions with a higher degree of financial marketization, due to the combined effects of lower levels of information asymmetry, higher levels of bank competition, and a richer array of corporate financing channels, the debt financing costs of local firms were already at a relatively low level before the implementation of the FCRS, with less room for reduction, hence the effect of FCRS in reducing debt financing costs is not evident.

5.3.4. Zombie firm governance

In China, the so-called “zombie firms” attracts much attention from academia and practitioners, the emergence of which is a result of administrative monopoly to some extent.⁶ Therefore, administrative monopoly regulation may help address the issue of zombie firms. Following Imai (2016), we categorize sample firms into normal firms (*Normal*) and zombie firms (*Zombie*). Columns (3) and (4) of Table 6 present the outcomes of the heterogeneity analysis taking into account the zombie firm governance, where the coefficient of *DID* in the *Normal* group is significantly negative, while the coefficient for the *Zombie* group, although negative, is not significant. This suggests that the FCRS can more significantly decrease the cost of debt of normal firms and weaken the advantages of zombie firms in obtaining financing from the banks. This helps to address the issue of zombie firms and mitigate their crowding-out effect on the available debt funding of normal firms. The FCRS restricts the government’s use of administrative power to support zombie firms, releasing credit resources that were previously occupied by zombie firms. Therefore, the implementation of FCRS helps to guide credit resources towards normal firms, resulting in a significant reduction in the cost of debt of normal firms.

5.3.5. Heterogeneity analysis based on review methods

The FCRS includes two modes of review: self-inspection and external supervision. We conjecture that the effectiveness of the policy’s implementation depends on the strictness with which local governments enforce these review methods. To examine this hypothesis, we conduct researches in two aspects: firstly, whether local governments have incorporated the fair-competition review into their performance evaluation system, that is, whether the rigor of self-inspection is strict or slack (Strict SI or Slack SI); secondly, whether local governments have commissioned third-party assessments, that is, whether the rigor of external supervision is strict or slack (Strict ES or Slack ES).⁷ The regression results, as displayed in Table 7, indicate that the coefficient of *DID* is both negative regardless of the rigor of self-inspection and external supervision, but the policy’s effect is more significant when the government conducts strict self-inspection and strict external supervision. Therefore, when strict self-examination and strict external supervision are combined, the FCRS can more effectively decrease the cost of debt.

6. Conclusion

As the anti-monopoly endeavors in the Asia-Pacific region continue to deepen, greater attention should be given to administrative monopoly, a special form of monopoly. The implementation of FCRS in China offers a valuable research opportunity for the academic community to study the policy effects of administrative monopoly regulation. This paper explores the influence of FCRS on corporate debt financing costs. The results show that the enforcement of FCRS significantly reduces corporate debt financing costs. The results of the parallel trend analysis, placebo check, PSM-DID approach, and other robustness tests suggest that our empirical results are reliable. Further research shows that FCRS can encourage firms to improve the accounting information quality, reduce corporate default risk, and thus lower the cost of debt. Moreover, the effect of FCRS in decreasing debt financing costs is more pronounced among private firms, small-sized and medium-sized firms, and firms in regions with lower degree of financial marketization. The FCRS also helps to address the issue of zombie firms and mitigate their crowding-out effect on the debt financing of normal firms. Furthermore, when the fair competition review measures include both strict self-inspection and strict external supervision, the effect of FCRS on reducing corporate debt financing costs is better.

Our research contributes to the literature on antitrust, administrative monopoly regulation, and corporate financing. Traditional antitrust research has mainly focused on market-based monopoly, with less attention given to monopolies formed by administrative power. This study introduces innovative research on administrative monopoly regulation, revealing that such regulation contributes to decreasing debt financing costs and highlighting the importance of anti-administrative monopoly. These findings also indicate that the

⁶ Zombie firms are defined as firms that, despite incurring losses or being insolvent, continue to operate. The key to the survival of zombie firms is the loans they receive from the governments or banks, which creates a crowding-out effect on the financing needs of normal firms.

⁷ The rigor of review methods implemented by Chinese provincial governments for the fair competition review is manually compiled by the author, with data primarily sourced from the webpage: https://www.sohu.com/a/201902118_123753.

Table 6

Heterogeneity analysis based on financial marketization level and zombie firm governance.

This table presents the results of heterogeneity analysis based on financial marketization level and zombie firm governance. Columns (1) and (2) respectively show the regression results for the high degree of financial marketization group (*High FML*) and low degree of financial marketization group (*Low FML*). Columns (3) and (4) respectively present the regression results for the normal firm group (*Normal*) and the zombie firm group (*Zombie*).

	(1)	(2)	(3)	(4)
	Financial marketization level		Zombie firm governance	
	Low FML	High FML	Normal	Zombie
Dep=	COD	COD	COD	COD
DID	−0.597*** (−3.174)	−0.439 (−1.599)	−0.452** (−2.607)	−0.299 (−0.884)
Control	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	4805	5324	9156	923
R-squared	0.690	0.703	0.679	0.804

Table 7

Heterogeneity analysis based on review methods.

This table presents the results of heterogeneity analysis based on review methods. The dependent variable for each column is the corporate cost of debt (*COD*). Columns (1) and (2) respectively present the empirical results for the strict self-inspection group (*Strict SI*) and the slack self-inspection group (*Slack SI*), while columns (3) and (4) present the results for the strict external supervision group (*Strict ES*) and the slack external supervision group (*Slack ES*).

	(1)	(2)	(3)	(4)
	Self-inspection		External supervision	
	Strict SI	Slack SI	Strict ES	Slack ES
	COD	COD	COD	COD
Dep=	(1)	(2)	(3)	(4)
DID	−0.391* (−1.878)	−0.268 (−1.448)	−0.407** (−2.292)	−0.167 (−1.040)
Control	Yes	Yes	Yes	Yes
Firm	YES	YES	YES	YES
Year	YES	YES	YES	YES
N	4768	5356	5531	4597
R-squared	0.733	0.685	0.717	0.695

FCRS is an effective measure for administrative monopoly regulation, contributing to the advancement of anti-monopoly efforts in the Asia-Pacific region.

CRediT authorship contribution statement

Shuxun Cai: Writing – original draft, Software, Investigation, Data curation. **Xiaoran Ni:** Writing – review & editing, Supervision, Methodology, Conceptualization.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pacfin.2025.102777>.

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